



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **RENEWCOILS ENGINEERING AND SUPPLY CO. LTD**
WPS No. **RNC-WPS-021** Date **23-May-13** Revision No. **0**
Supporting PQR No.(s) **RNC-PQR-021** Date **29-Apr-13** Type(s) **Manual**
Welding Process(es) **Gas Tungsten Arc Welding**
(GTAW)

JOINT (QW-402)

Joint Design **GROOVE & FILLET**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal

☐ Nonmetallic ☐ Other

Notes _____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **5B** Group No. **1** to P-No. **5B** Group No. **1**

Material Specification Type and Grade **A335 P5, F182-F5, A234 WP5** TO **A335 P5, F182-F5, A234 WP5**

Chem. Analysis and Mech. Prop. **N/A**

Thickness Range :

Base Metal : Groove **1.6-14.22** Fillet **ALL**

Pipe Dia. Range : Groove **ALL** Fillet **ALL**

Other **N/A**

FILLER METALS (QW-404)

	GTAW	SMAW
Spec. No. (SFA)	5.28	
AWS No. (Class)	ER80S-B6	
F-No.	6	
A-No.	5	
Size of Filler Metals	Ø 2.4 ~ 3.2mm.	
Deposited Weld Metals	7.11 mm.	
Thickness Range :		
Groove	14.22 mm.	
Fillet	ALL	
Electrode-Flux (Class)	-	
Flux Trade Name	-	
Consumable Insert	-	
Other	BOHLER CM 5-IG or Equivalent	

POSITION (QW-405)

Position(s) of Groove **6G**

Welding Progression Up ☒ X Down ☐

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **714°C~770°C ±10°C**

Time Range **Holding Time 2 Hour**

Notes **Heating Rate = 200°C Max/Cooling Rate = 200°C Max**

PREHEAT (QW-406)

Preheat Temp. Min. **250°C**

Interpass Temp. Max. **300°C**

Preheat Maintenance **Continuous at 300 °C 1 HR.**

Notes **Resistance heater**

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	12-20 Ltr/Min
Trailing	-		-
Backing	Argon	99.99%	13-20 Ltr/Min

WPS No.

RNC-WPS-021

Rev.

0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☐ AC ☒ DC Polarity EN(GTAW)
Amps (Range) See Below Volts (Range) See Below
Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
(Pure Tungsten, 2% Thoriated, etc.)
Mode of Metal Transfer for GMAW N/A
(Spray arc, short circuiting arc, etc.)
Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
Orifice or Gas Cup Size 8.0 mm - 19.0 mm
Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
Method of Back Gouging N/A
Oscillation N/A
Contact Tube to Work Distance N/A
Multiple or Single Pass (per side) MULTIPASS
Multiple or Single Electrodes SINGLE ELECTRODE
Travel Speed (Range) See Below
Peening N/A
Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1 st	GTAW	ER80S-B6	2.4 mm.	DC EN	80-110	8-12	4-7	
2 nd and Other	GTAW	ER80S-B6	2.4 mm.	DC EN	100-130	10-13	5-9	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME-IX, 2010 latest Revision.

Prepared by:

Mr. Kittipong M.
Name

Signed

Date

Reviewed by:

ภูชิน สีหอมไชย
Phuchin Seehomchai
Name

Signed

Date

Approved by:

ภูชิน สีหอมไชย
Phuchin Seehomchai
Name

Signed

Date

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number: RNC-PQR-021

Tensile Test Report No.TS-13-05-049

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
1263/13 TR1	19.37	6.82	132.103	64.214	486.090	OUT OF WELD
1263/13 TR2	19.36	6.79	131.454	64.567	491.176	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-05-008

Type and Figure Number	Type of Bend	Indication	Results
1263/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
1263/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
1263/13 BR1	Transverse root bend	No Open Discontinuity	Accepted
1263/13 BR2	Transverse root bend	Open Discontinuity 1.0 mm	Accepted

Toughness Tests (QW-170) Report No.IM-13-05-016

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Average (Joules)	Lateral Expansion Mils
1263/13-1	Weld Metal	V-Notch	- 20 °C	117.9	120.53	
1263/13-2	Weld Metal	V-Notch	- 20 °C	119.69		
1263/13-3	Weld Metal	V-Notch	- 20 °C	124.01		
1263/13-1	HAZ	V-Notch	- 20 °C	186.39	181.19	
1263/13-2	HAZ	V-Notch	- 20 °C	189.53		
1263/13-3	HAZ	V-Notch	- 20 °C	167.64		
1263/13-1	Base Metal	V-Notch	- 20 °C	142.46	108.93	
1263/13-2	Base Metal	V-Notch	- 20 °C	86.96		
1263/13-3	Base Metal	V-Notch	- 20 °C	97.36		

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR-010

Hardness Testing Report No. HV-13-05-003

Macrostructure Report No. MA-13-05-001

Welders Name Mr.Amnaj N.

Welder Number

W-061

Tests Conducted By Mr.Keerati Sa-ngoen

Laboratory Test

LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

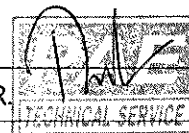
Manufacture or Contractor : _____

Inspected by : Mr.Wanlop N.

Date : _____

Certified by : Mr.Chaloemkiet R.

Date : 20 May 2013



WELDING PROCEDURE

PQR No. : RNC-PQR-021

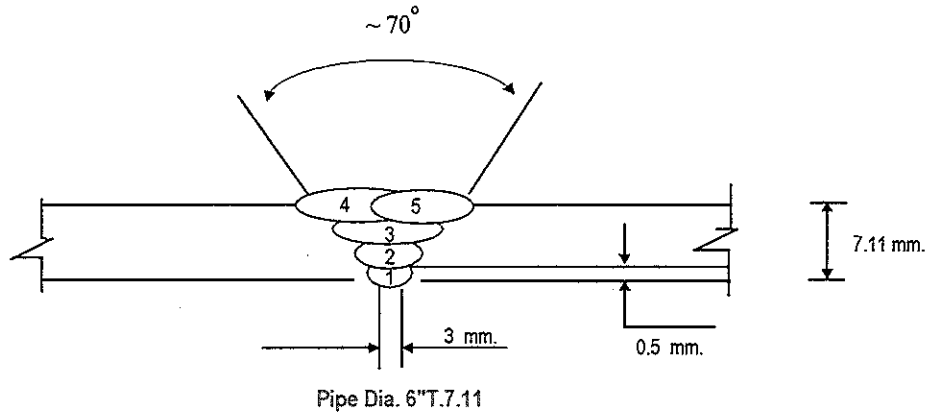
WPS No. : RNC-WPS-021

QUALIFICATION RECORDS

THIS PQR RECORDS ACTUAL TEST

CONDITION AND RESULT

Welding Sequences for RNC-PQR-021 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave Bead	Wire Feed Speed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
1	GTAW	ER80S-B6	2.4	80	9.5	DCEN	-	-	5.50	8.29	15	17
2	GTAW	ER80S-B6	2.4	130	9.5	DCEN	-	-	6.20	11.95	15	17
3	GTAW	ER80S-B6	2.4	130	10.2	DCEN	-	-	6.30	12.60	15	17
4	GTAW	ER80S-B6	2.4	130	11	DCEN	-	-	6.00	14.00	15	17
5	GTAW	ER80S-B6	2.4	130	11	DCEN	-	-	6.90	13.20	15	17



PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 682208-9

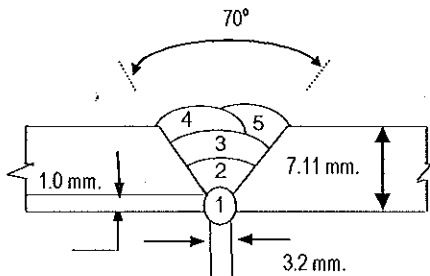
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : MINI SHUTDOWN PTTGC		Location : RNC Work Shop		Report No. PQR-008	
Customer Order No. : N/A		WPS No. : RNC-WPS-021		Date : 29-Apr-13			

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-061	MR.ANANJ N.	GTAW	6G	ASME IX	A335 P5	ø 6" Thk. 7.11 mm.	ER80S-B6	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER80S-B6	3.2	DCEN	80	9.5	5.5	205	8.3	Argon	15	Argon	17
2	GTAW	ER80S-B6	3.2	DCEN	130	9.5	6.2	207	12.0	Argon	15	Argon	17
3	GTAW	ER80S-B6	3.2	DCEP	130	10.2	6.3	212	12.6	Argon	15	Argon	17
4	GTAW	ER80S-B6	3.2	DCEP	130	11	6.0	215	14.0	Argon	15	Argon	17
5	GTAW	ER80S-B6	3.2	DCEP	130	11	6.9	215	13.2	Argon	15	Argon	17

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

Mr. Bayam S.

30-Apr-2013

SIGNATURE :

NAME :

DATE :

Mr. Phuchin S.

30/05/13

SIGNATURE :

NAME :

DATE :

Phuchin Seehomchai

Phuchin Seehomchai



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnplm.co.th



Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-05-049

Revision No. 0

Test Product	PQR No. RNC-PQR-021	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P5	Type of Test	Reduced Section Tension
Dimension	Ø 6" #40 (7.11 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	6 May 2013	Equipment/ Serial No.	SHIMADZU UH200A / 32098926
Date Tested	8 May 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	1263/13 TR1	N/A	6.82	19.37	132.103	N/A	N/A	N/A	64.214	486.090	N/A	OUT OF WELD
2	1263/13 TR2	N/A	6.79	19.36	131.454	N/A	N/A	N/A	64.567	491.176	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-018

Welding Process : GTAW

Filler/Electrode Spec't : ER80S-B6

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 20 / 05 / 13

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 20 / 05 / 13

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30454

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-05-008

Revision No. 0

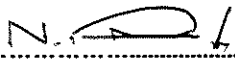
Test Product	PQR No. RNC-PQR-021	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P5	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 6" #40 (7.11 mm Thk.)	Diameter of Meudrel	28 mm
Date Received	6 May 2013	Bend Angle	180 Degree.
Date Tested	8 May 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	1263/13 BF1	Transverse face bend	7.25x37.95x162.18 mm	No Open Discontinuity	N/A	-
2	1263/13 BF2	Transverse face bend	7.25x38.03x163.48 mm	No Open Discontinuity	N/A	-
3	1263/13 BR1	Transverse root bend	7.20x38.06x163.47 mm	No Open Discontinuity	N/A	-
4	1263/13 BR2	Transverse root bend	7.21x37.92x162.78 mm	Open Discontinuity	WELD ZONE	1.0 mm

Additional details : WPS No. RNC-WPS-018

Welding Process : GTAW

Filler/Electrode Spec't : ER80S-B6

Reported by : 

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 20 / 05 / 13

Approved by : 

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 20 / 05 / 13

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0457

TESTING
LABORATORY

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-05-016

Revision No. 0

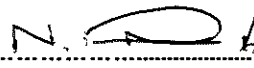
Test Product	PQR No. RNC-PQR-021	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ⁰¹
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P5	Type of Test	Charpy V-Notch Impact
Dimension	Ø 6" #40 (7.11 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	6 May 2013	Equipment 's Capacity	542 Joules
Date Tested	8 May 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	1263/13-1	-20°C	5.008	10.009	54.84	186.39	181.19	HAZ	-
2	1263/13-2	-20°C	5.010	10.012	54.94	189.53			
3	1263/13-3	-20°C	5.011	10.013	54.89	167.64			
4	1263/13-1	-20°C	5.009	10.017	54.87	142.46	108.93	Base Metal	-
5	1263/13-2	-20°C	5.010	10.017	54.88	86.96			
6	1263/13-3	-20°C	5.009	10.013	54.89	97.36			
7	1263/13-1	-20°C	5.013	10.010	54.91	117.90	120.53	Weld Metal	-
8	1263/13-2	-20°C	5.014	10.014	54.96	119.69			
9	1263/13-3	-20°C	5.011	10.011	54.97	124.01			

Additional details : WPS No. RNC-WPS-018

Welding Process : GTAW

Filler/Electrode Spec't : ER80S-B6

Reported by : 
 (Mr. Keerati Sa-ngoen)
 (Acting) General Manager - Quality Assurance
 Date / / 2013

Approved by : 
 (Mr. Pavaret Freedawiphat)
 Deputy Managing Director
 Date / / 2013

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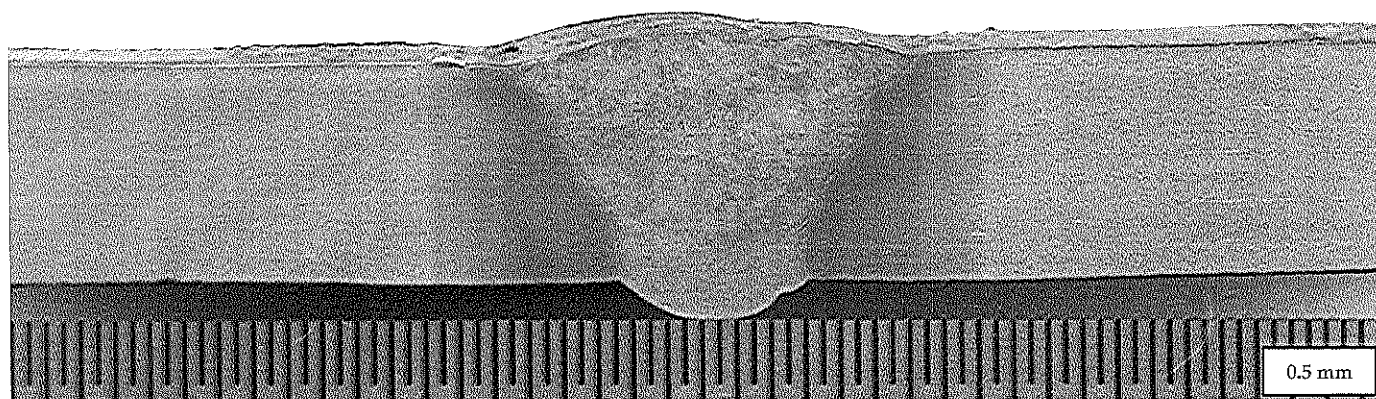
Macrostructure Analysis Report**Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.****Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.**

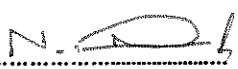
Page 1 of 1

Report No. MA-13-05-001

Revision No.0

Test Product	PQR No.RNC-PQR-021	Material Characterization	Pipe to Pipe
Test No.	1263/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A335P5	Method of Preparation	ASME Section IX : 2010 edition
Dimension	Ø6" #40 (7.11 mm Thk.)	Test Method	ASME Section IX : 2010 edition
Welding Process/Position	GTAW	Standard of Test	ASME Section IX : 2010 edition
Type of Joint	Butt Joint	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	N/A	Time of Etching	5 minutes
Date Received	6 May 2013	Date Tested	6 May 2013

**Macro-etch Examination :** The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.**Additional details : -**

Reported by : 
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 6 May 2013

Approved by : 
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 6 May 2013

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13329

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

Hardness Testing Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Page 1 of 1

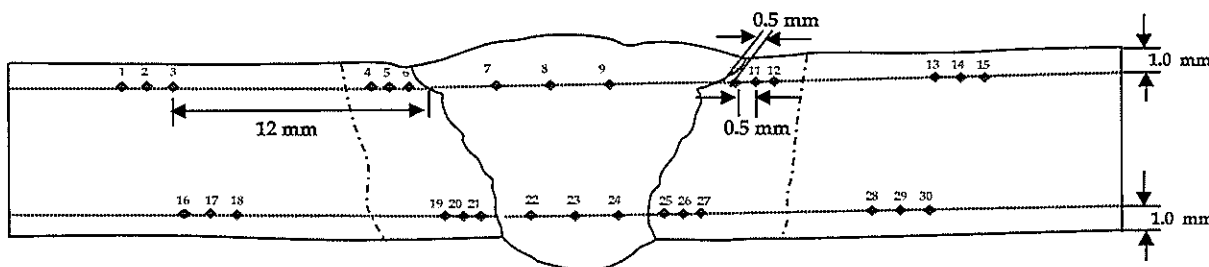
Report No. HV-13-05-003

Revision No. 0

Test Product	PQR No.RNC-PQR-021	Material Characterization	Pipe to Pipe
Test No.	1263/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ^{E2}
Project Name	N/A	Standard of	ASME section IX: 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A335P5	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	Ø6" #40 (7.11 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	6 May 2013	Cal. Block	No. A60023 No. A62072
Date Tested	6 May 2013	Ambient Temperature	27.3 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)
1	BASE	133.1	16	BASE	136.1	*****		
2	BASE	131.2	17	BASE	135.8			
3	BASE	133.7	18	BASE	136.5			
4	HAZ	187.7	19	HAZ	196.4			
5	HAZ	203.9	20	HAZ	209.0			
6	HAZ	206.2	21	HAZ	214.4			
7	WELD	196.0	22	WELD	194.7			
8	WELD	195.2	23	WELD	193.9			
9	WELD	200.7	24	WELD	204.5			
10	HAZ	193.8	25	HAZ	209.2			
11	HAZ	191.4	26	HAZ	210.9			
12	HAZ	137.2	27	HAZ	206.4			
13	BASE	130.6	28	BASE	131.6			
14	BASE	132.2	29	BASE	131.8			
15	BASE	130.8	30	BASE	131.8			

Additional details : -

Reported by :
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 20/05/13

Approved by :
(Mr.Pavaret Freedawiphat)
Deputy Managing Director
Date : 20/05/13

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GLOBAL HEAT TREATMENT (THAILAND) CO., LTD.

PREHEAT REPORT

CLIENT:	RENEW COILS ENGINEERING AND SUPPLY CO.-LTD.		
PROJECT:	MINI SHUT DOWN PTTGC 2013		
CONTRACTOR:	GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.		
PREHEAT METHOD:	☐ GAS	☒ ELECTRIC	COMPLETION DATE: 29-04-2013
LOCATION:	☒ SHOP	☐ FIELD	REPORT NO. PH-003
RECORDER NO:	EH- 07ZA194		PAGE NO. 1
CHART SPEED:	☒ 25 MM/Hr	☐ 50 MM/Hr	☐ 100 MM/Hr

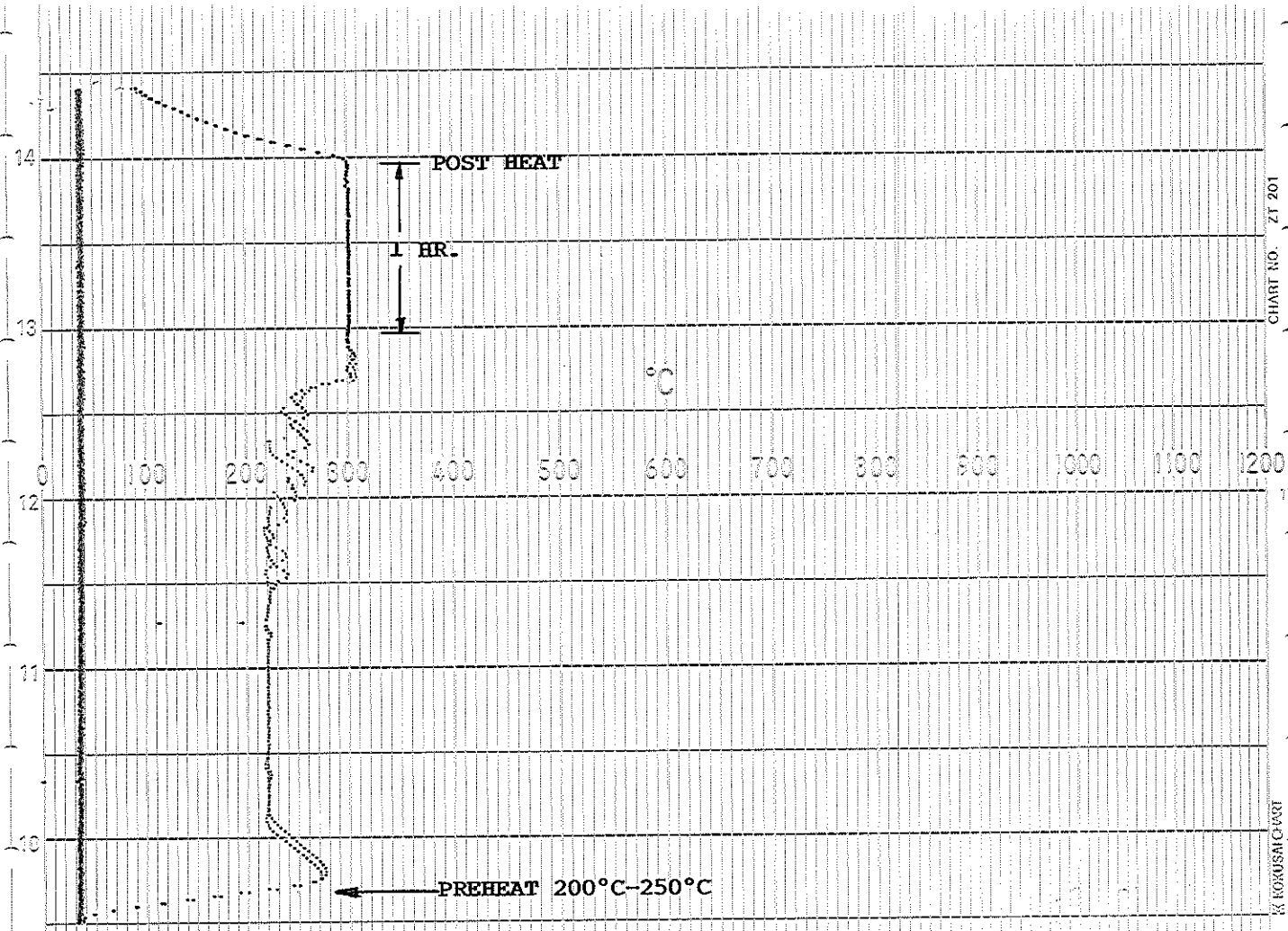
RESULT RECORDER CHART TO BE ATTACHED

DESCRIPTION	REQUEST	ACTUAL
1) PREHEAT TEMP (C° MIN)	200-250	220
2) INTERPASS TEMP (C° MAX)	300	265
3) POSTHEAT TEMP (C°)	300	300
4) POSTHEAT TIME (Hr)	1 HR.	1 HR.

MATERIAL : A335 P5

[illegible]

CHECKED BY (GLOBAL HEAT TREATMENT THAILAND)	APPROVED BY	APPROVED BY
SIGN:	SIGN:	SIGN:
NAME: KOMKHUN D.	NAME: Mr. Chatchawan S.	NAME: ภูชิน สีหอนไชย
DATE: 29-04-2013	DATE: 30/04/13	DATE: Phuchin Seehomcha



DATE. 29-04-2013

CHART NO. PH - 003

RECORDER NO. EH 07ZA194

CHART SPEED. 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

PROJECT. MINI SHUT DOWN PTTGC 2013

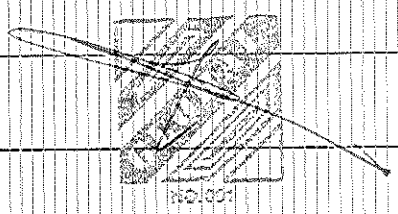
LINE NO. RNC-WPS-021

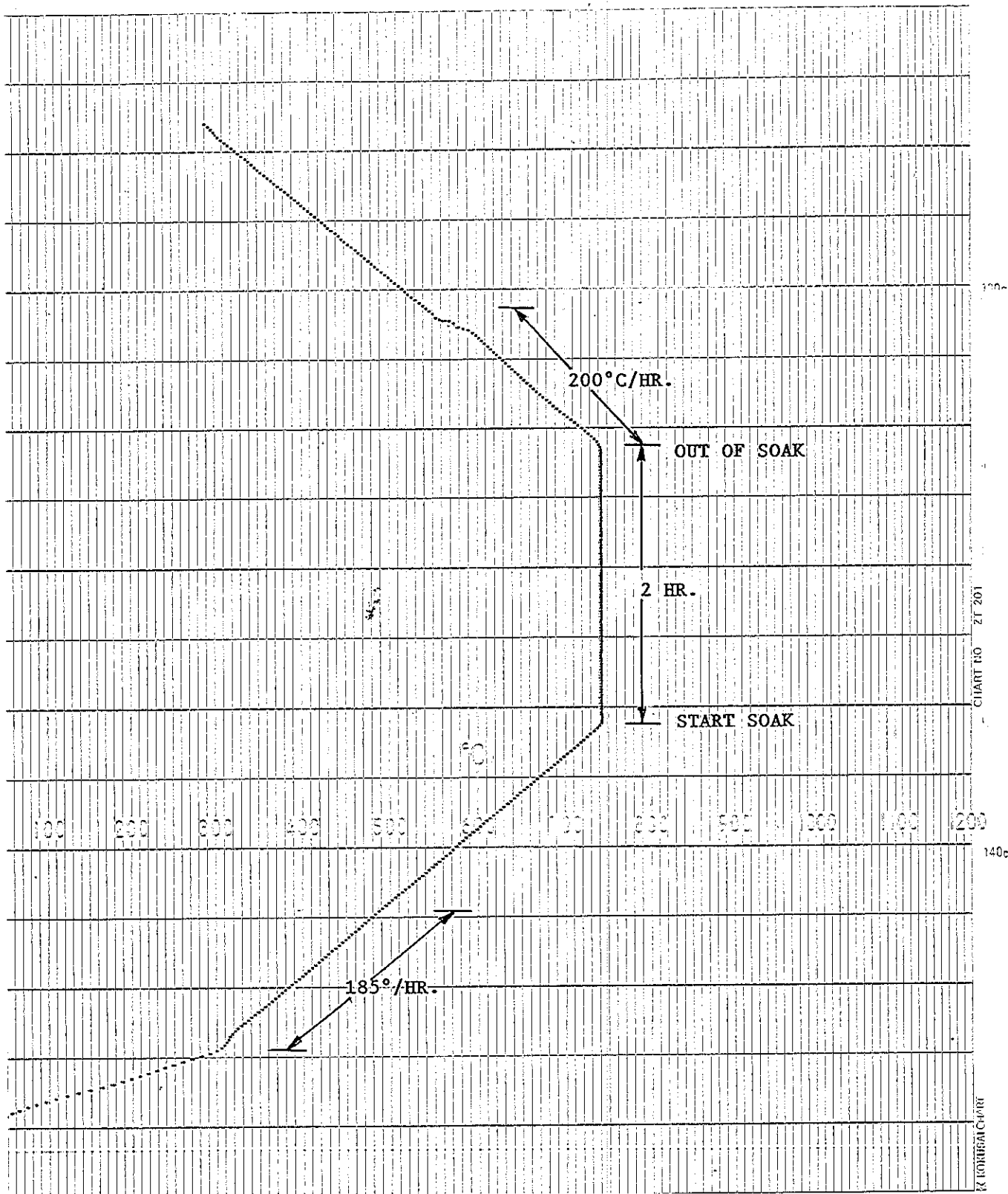
SIZE. 6" THK. 7.11 mm.

MATERIAL. A335 P5

START TIME. 11.30 AM.

TECHNICIAN. MR. CHALERMCHAI NG.





DATE. 30-04-2013

CHART NO. PW - 003

IC	COLOUR
1.	RED
2.	BLACK

RECORDER NO. EH 07ZA194

CHART SPEED. 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO. LTD.

PROJECT. MINI SHUT DOWN PTTGC 2013

LINE NO. RNC-WPS-021

SIZE 6" THK. 7.11 mm.

MATERIAL. A335 P5

START TIME. 09.20 AM.

TECHNICIAN. MR. PHANOMTUAN CH.



NO. 0021

500.0153

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)682208-9, Fax : 038)682206, SRIRACHA : 038)330220, 330229

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Abnahmeprüfzeugnis 3.1

Inspection certificate 3.1

nach / as per: EN 10204

Nr. / No.: 2-2012-25-2198182

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Bestell-Nr.	Order No.	email	vom / of	15.12.2011	100049															
Auftrags-Nr.	Works order	1002118357																		
Lieferschein/Pos./Split	Dispatch note/pos./Split	2002203341 / 0020 / 900003	vom / of	15.12.2011																
Prüfgegenstand	Test object	Schweißstab welding rod			10621															
Handelsname	Trade designation	BOEHLER CM 5-IG			X373EW02															
Produktkennzeichnung	Marking of product																			
Normbezeichnung	Standard classification	EN ISO 21952-A - W CrMo5 Si / AWS A5.28-05: ER80S-B6																		
Abmessung	Dimension	2.40 x 1000 mm (.094" x 39")																		
Charge	Heat No.	145351																		
Liefermenge	Quantity delivered	405 KG (893 LB)																		
Anforderungen	Requirements																			
Chemische Analyse in % Chemical composition in %										Stab rod										
Charge	C	Si	Mn	P	S	Cr	Mo	Ni	V	Cu	Al									
Heat No.																				
145351	0.070	0.45	0.52	0.010	0.005	5.67	0.56	0.08	0.010	0.07	0.005									
Zugversuch Tensile Test	nach according to	ASTM E8						Probenvorbereitung Specimen preparation									AWS B4.0			
	Prüftemp. T Test temp. °C	Streckgrenze Reh Yield point MPa		Dehngrenze Rp Yield strength 0.2% MPa 1.0%		Zugfestigkeit Rm Tensile strength MPa		Dehnung A Elongation % Lo=4d		Einschnr. Z Reduction %		Bemerkung Remarks								
Minimum	20			470		550		17				730-760°C/1h								
Maximum																				
Kerbschlagv. Impact Test	nach according to	ASTM E23						Probenvorbereitung Specimen preparation									AWS B4.0			
	Prüftemp. T Test temp. °C	Kerbschlagarbeit Mindest. KV Absorbed energy minimum values J				Mittelwert Average J		Lat. Breitg. Lateral expansion mm		Krist. Bruchanteil Shear fracture %		Bemerkung Remarks								
	20°C (68°F)	47										730-760°C/1h								

Bemerkung
Remarks The product BOEHLER CM 5-IG meets the requirements of the filler metal specification ASME, sec.II, part C, AWS A5.28-05, classification ER80S-B6 when tested in accordance with that specification. Produced according to AWS A5.01, class S3

Ort / Town
Kapfenberg

Datum / Date
02.01.2012

Dieses Zeugnis wurde maschinell erstellt und gilt ohne Unterschrift.
This certificate was issued by DP-equipment and does not require signature.

Abnahmebeauftragter
Authorized representative
Winkler

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ATU 27116607
DVR 0862348

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Swiftcode: DEUTAT33
IBAN: AT37 1910 0000 3148 6000